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BELAGAVI, KARNATAKA.

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ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A MINI PROJECT REPORT

ON

“AGRO MARKETING AND MANAGING SYSTEM”

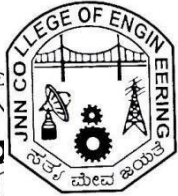
Submitted to Visvesvaraya Technological University in partial fulfillment of the requirement for the award of Bachelor of Engineering degree in Computer Science and Engineering.

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CERTIFICATE

This is to certify that the mini project is entitled

“AGRO MARKETING AND MANAGING SYSTEM”

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ABSTRACT

The main aim of developing “Agro Marketing and Managing System” application is to help farmers by providing all kinds of agriculture related information in the site. “Agro Marketing and Managing System” is web application which helps farmers to share best-practice farming processes. It helps farmers to improve their productivity and profitability. It enables farmers to sell their products online and farmers can purchase tools and seeds directly from seller. Farmers can view their profile, and they can register, edit and delete data.

The farmers can sell their productions online and the buyer can purchase various agricultural products online. Buyers can send purchase request to check the quality of the Agro product through mails

ACKNOWLEDGEMENT

At the very outset of this report on “An encryption approach to achieve Secure Data Transfer”, we would like to extend our sincere & heartfelt obligation towards all the persons who have helped us in this endeavor. Without their active guidance, help, cooperation & encouragement, we would not have made headway in the project.

We are grateful to our institution **JNN College of Engineering** and the Department of Computer Science and Engineering for imparting to us the knowledge with which we can do our best. We would like to thank our beloved guide **Dr. Poornima K M**, Professor, Dept. of Computer Science and Engineering and our coordinator **Dr. Benakappa S M** Associate professor, Dept. of Computer Science and Engineering.

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CHAPTER 1

INTRODUCTION

Agriculture is the backbone of many economies, providing food, raw materials, and employment to a significant portion of the population. However, despite its critical importance, the agricultural sector faces numerous challenges, particularly in marketing and managing agricultural produce efficiently. Farmers often struggle with limited access to markets, unfair pricing, and inefficient supply chain management. These challenges result in financial instability, wasted resources, and reduced profitability. The Agro Marketing and Managing System is designed to address these issues by providing a comprehensive, technology-driven platform that bridges the gap between farmers, buyers, and market opportunities. It integrates modern digital tools to streamline the marketing, inventory management, and logistics processes involved in agriculture. The Agro Marketing and Management System (AMMS) is a transformative framework designed to modernize and streamline the processes of marketing, managing, and distributing agricultural products. Agriculture, being one of the most critical sectors for global sustenance and economic growth, faces numerous challenges, including fragmented markets, lack of real-time data, inefficient supply chains, and exploitation by intermediaries. The AMMS addresses these challenges by leveraging technology to create a seamless interface for farmers, buyers, distributors, and policymakers, ensuring a more equitable and efficient agricultural ecosystem. At its core, the AMMS acts as a bridge between farmers and markets, facilitating better access to buyers and ensuring fair pricing mechanisms. Traditional agricultural practices often result in farmers receiving limited information about market trends, prices, and demand for their products, which leaves them vulnerable to exploitation. Through the integration of real-time market data, AMMS provides farmers with the tools and knowledge they need to make informed decisions about when, where, and how to sell their produce. This real-time data includes up-to-date market prices, demand forecasts, and seasonal trends, enabling farmers to optimize their profits and reduce waste. One of the standout features of the AMMS is its ability to streamline the supply chain. The agricultural supply chain is often riddled with inefficiencies such as delays in transportation, improper storage, and excessive reliance on middlemen. AMMS incorporates advanced logistics and inventory management systems to address these inefficiencies. By tracking the movement of goods from the farm to the market, the system ensures timely delivery, reduces post-harvest losses, and improves overall product quality.

1.1 PROBLEM STATEMENT

In the agricultural sector, farmers often face challenges related to marketing their products, managing their operations, and reaching the right buyers. The traditional methods of selling agricultural products can be inefficient, with limited access to market opportunities, delayed payments, and poor transparency in pricing. Additionally, farmers often struggle to manage inventory, track sales, and ensure timely delivery. This leads to financial instability, lack of fair pricing, and the inability to leverage new market trends. The Agro Marketing and Managing System (AMMS) aims to solve these challenges by providing an integrated platform that connects farmers with buyers, provides tools for efficient inventory and supply chain management, and offers insight into market trends. The system will enable farmers to: Market their Products Efficiently, Manage Inventory, Ensure Fair Pricing and Transparency, Streamline Sales and Transactions, Optimize Delivery Logistics, Gain Insights for Decision Making. The goal of the Agro Marketing and Managing System is to enhance the efficiency and profitability of the agricultural supply chain by bridging the gap between farmers and markets, improving resource management, and promoting financial stability for farmers.

1.2 OBJECTIVES OF THE PROJECT

- The main objective of the project is to design and develop a user-friendly system and it also an efficient computerized system and based on GUI
- To provide synchronized and centralized farmer, seller database and it will eliminate data redundancy.
- Provides Security to the data by using login and password method.
- To provide immediate storage and retrieval of data and information.

1.3 SCOPE OF THE PROJECT:

- Real-time market accessibility
- Geographic scope
- Direct market linkages
- Stakeholders
- System functionality
- System Functionality
- User Roles and Accessibility
- Technology Integration

1.4 ORGANIZATION OF THE REPORT

This section deals with the introduction and organization of the project report.

- ✓ Chapter 1: Gives information about the introduction of our topic.
- ✓ Chapter 2: Discuss about the study of existing system.
- ✓ Chapter 3: Discuss about the domain specific
- ✓ Chapter 4: Discuss about the System Design and Implementation.
- ✓ Chapter 5: Gives information about the snapshot and results.

And the end of the report, conclusions and future enhancement and bibliography is included

CHAPTER 2

STUDY OF EXISTING SYSTEM

2.1 CASE STUDY

Source trace is collaborating with small farmers agri-business consortium and Karnataka horticulture department, deploying its digital solutions to support the horticulture farmers of India. Karnataka agriculture department is committed to providing a responsive and effective mechanism for the welfare of farmers and farm-based communities and recognizes the need to harness the growing power of information technologies for the betterment of the life of the farmers and management of farmer producer organizations in Haryana. To deploy its digital solution, source trace is in the process of creating 100,000 farmer profiles. The system was developed using technologies such as html, css, js and mysql. Python- flask, html and css are used to build the user interface and database was built using mysql. The system is free of errors and very efficient and less time-consuming due to the care taken to develop it. All the phases of the software development cycle are employed, and it is worthwhile stating that the system is very robust. Provision is made for future development in the system.

2.2 CASE EXAMPLE

FARMER STORY:

Ramesh, a small-scale vegetable farmer, previously sold his crops to a local agent at a low price. After joining the AMMS platform:

- Ramesh listed his fresh tomatoes on the marketplace at ₹20/kg (against the agent's offer of ₹12/kg).
- Within 24 hours, a buyer from the nearest city placed an order for 500 kg.
- Using the platform's logistics service, the tomatoes were transported directly to the buyer.
- Ramesh earned ₹10,000 instead of ₹6,000, significantly boosting his income.

Key Metrics		
Metric	Before AMMS	After AMMS
Average Farmer Income	₹15,000/month	₹20,000/month
Wastage During Transit	20%	5%
Buyer Satisfaction Rate	70%	90%
Farmer Registration	N/A	5,000+ farmers within the first 6 months

Fig 2.1: key metrics

2.3 PROPOSED SOLUTION

The farmers can sell their productions online and the buyer can purchase various agricultural products online. Buyers can send purchase requests to check the quality of the product. After collecting all the farm produce from the farmers, it should be sold to the customers. This project covers these entries and the data collections. There are 2 types of users: Customer & Farmers. The login id and password must be required to login the system. The article and agro products section help farmers to share their products and increase profitability.

1. FARMER MODULE:

- **Farmer Registration and Profile:** Registration with details such as crops grown, land size, and contact info.
- **Product Listing:** Add products for sale, including quantity, price, and expected harvest date.
- **Inventory Management:** Monitor stored products and manage availability.
- **Price Updates:** Real-time pricing data for crops in various markets.
- **Knowledge Hub:** Access to agricultural tips, weather updates, and expert advice.

2. BUYER MODULE:

- **Buyer Registration and Profile:** Register buyers, including wholesalers, retailers, and exporters.
- **Product Search and Filter:** Search for specific crops by category, location, or availability.

- **Order Placement:** Place orders directly with farmers.
- **Pricing Negotiation:** A chat or bidding system for price discussions.

3. MARKETPLACE:

- **Dynamic Pricing System:** Ensures fair pricing based on demand and supply.
- **Market Trends:** Provide data on high-demand crops and upcoming market needs.
- **Feedback and Ratings:** System for buyers and farmers to rate each other.

4. LOGISTICS AND SUPPLY CHAIN MANAGEMENT:

- **Logistics Coordination:** Connect farmers with logistics providers for transport.
- **Tracking System:** Track shipments from farm to buyer.
- **Warehouse Management:** Integration with warehouses for temporary storage.

5. ADMIN MODULE:

- **User Management:** Oversee registration and resolve disputes.
- **Market Monitoring:** Track prices and trends to prevent exploitation.
- **Advertisement Management:** Enable third-party ads or promotions.
- **Analytics Dashboard:** Insights on transactions, user engagement, and market performance.

2.4 KEY BENEFITS

- **Empowers Farmers:** Provides direct access to buyers and better market information.
- **Reduces Middlemen:** Streamlines transactions, reducing unnecessary costs.
- **Transparency:** Ensures fair pricing and reduces exploitation.
- **Data-Driven Decisions:** Market insights help farmers and buyers make informed decisions.
- **Farmer Module:** Enables farmers to list their produce and access real-time market data.
- **Buyer Module:** Allows buyers to search for products, negotiate prices, and place orders.
- **Marketplace:** Facilitates transactions, bidding, and payment processing.
- **Knowledge Hub:** Provides agricultural tips and expert advice.

2.5 CHALLENGES ADDRESSED

- **Lack Of Market Access:** Farmers previously sold their produce through intermediaries, earning less profit. With AMMS, they list their crops directly in the marketplace, connecting with buyers from urban areas.
- **Logistics Issues:** Many farmers struggled with transporting their goods to markets. AMMS integrated local logistics providers, enabling farmers to book transport services directly.
- **Price Transparency:** Farmers often sold their produce at prices dictated by middlemen. AMMS provided real-time market prices, ensuring that farmers set competitive yet fair rates.
- **Knowledge Gap:** AMMS provided a knowledge hub that offered weather forecasts, pest management tips, and tutorials on sustainable farming practices.

2.6 METHODOLOGY

- **Improved agricultural land management:** A methodology that measures the reduction of greenhouse gas emissions and soil organic carbon removal from improved agricultural practices. These practices include reducing fertilizer and tillage, improving water management, and improving grazing practices
- **Farm management:** A decision-making process that involves evaluating and implementing different farming strategies. Farm management methods help farmers make decisions by analyzing resource use and presenting the consequences of different decisions.
- **Seedbed preparation:** A process that involves preparing the soil for sowing seeds. This includes tilling, levelling, digging, and fertilizing the soil. A well-prepared seedbed can help with germination, crop survival, and ensure proper space and sunlight for the crops.
- **Drainage:** A process that involves draining away stagnant water and lowering the water table to a depth that allows crops to grow. In tropical areas, irrigation can supplement drainage facilities to maintain water levels during droughts.
- **Reclamation:** A process that involves reclaiming small swamps on farms or groups of farms. Public bodies usually reclaim large swamps

CHAPTER 3

DOMAIN SPECIFIC

3.1 SOFTWARE AND HARDWARE REQUIREMENTS SPECIFICATION

❖ Software requirements:

Frontend- HTML, CSS, Java Script, Bootstrap

Backend-Python flask (Python 3.7) , SQL Alchemy,

- Operating System: Windows 10
- Google Chrome/Internet Explorer
- XAMPP (Version-3.7)
- Python main editor (user interface): PyCharm Community
- workspace editor: Sublime text 3

❖ Hardware requirements:

- Computer with a 1.1 GHz or faster processor
- Minimum 2GB of RAM or more
- 2.5 GB of available hard-disk space
- 5400 RPM hard drive
- 1366 × 768 or higher-resolution display
- DVD-ROM drive

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 SYSTEM ARCHITECTURE:

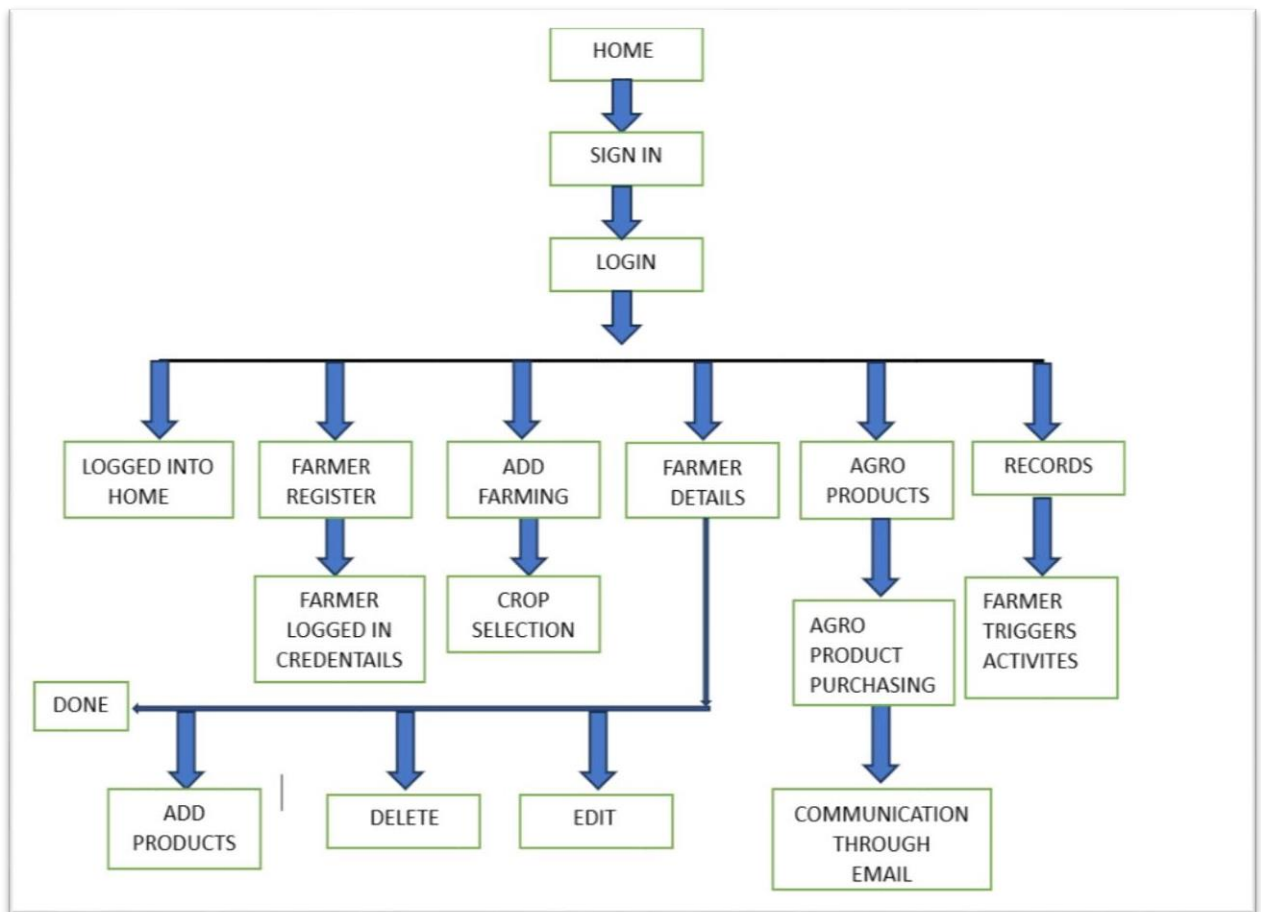


Fig 4.1: System Architecture flow diagram

The provided flowchart outlines the operational workflow of a comprehensive agricultural management system designed to streamline user interactions, farming processes, and product management. The system begins at the Home page, offering users two primary entry points: Sign In for new users and Login for existing users. Once authenticated, users are directed to the Logged-In Home, which acts as the central hub, presenting various modules tailored to meet specific agricultural and user management needs.

The Farmer Register module allows users to register farmers and manage their credentials systematically. This ensures that detailed records of farmers, including their personal and

farming-related data, are securely stored and accessible for future use. After registration, these farmers' details are managed under Farmer Logged-In Credentials, enabling efficient tracking and updates to their information.

The Add Farming section focuses on the farming aspect, allowing users to specify their farming activities through Crop Selection. This feature is crucial for categorizing and managing various farming types, helping users better organize and implement agricultural operations based on specific crops or practices.

The Farmer Details module is a versatile component where users can perform a variety of actions, including Add Products, Edit, and Delete records related to farmers. This feature enhances the flexibility and accuracy of managing farmer information, ensuring that any changes or updates can be applied seamlessly. The Add Products option integrates with the Agro Products section, enabling users to list agricultural products for sale or distribution.

The Agro Products module extends into Agro Product Purchasing, facilitating the buying and selling of agricultural goods. This feature is designed to connect farmers and buyers, promoting efficient product marketing and enhancing the visibility of agro-products in the market. Additionally, the module emphasizes product promotion to attract more buyers and ensure successful transactions.

The Records module serves as a log for various system activities and interactions. It tracks Farmer Trigger Activities, which include key actions initiated by farmers, and maintains a robust system of Communication Through Email. This communication feature ensures that farmers and users are kept informed about updates, transactions, and other essential information, fostering effective collaboration.

Overall, the flowchart provides a well-structured framework for managing the core functionalities of an agricultural system. It enables users to navigate through farmer registration, farming activities, product management, and communication with ease. By integrating these components into a single workflow, the system enhances productivity, streamlines operations, and facilitates effective communication between all stakeholders. Its modular design ensures scalability and adaptability to diverse agricultural requirements, making it a practical tool for managing both small-scale and large-scale farming operations. This flowchart ultimately represents an efficient solution for bridging gaps in agricultural management and fostering a more connected and resourceful farming community.

4.2 E-R DIAGRAM:

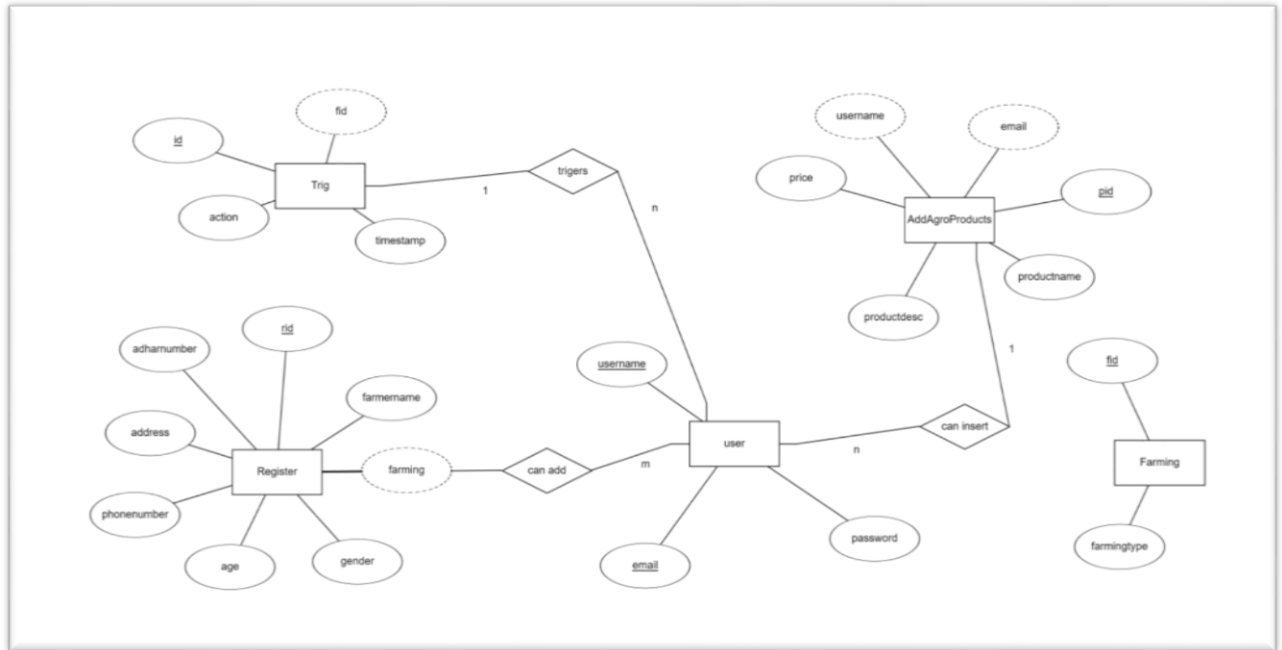


Fig 4.2: E R Diagram

The image represents an Entity-Relationship (ER) diagram, which is a conceptual model used to design a database structure. It depicts entities, their attributes, and the relationships between them. The system appears to be for managing agricultural data, involving farmers, users, farming types, and agricultural products. Here's a detailed explanation:

The Register entity stores farmer-specific details, including Aadhaar number, address, phonenumber, age, gender, and farmername, with rid as the primary key. Farmers are associated with the User entity through the "can add" relationship, which is one-to-many (1:M), indicating that one farmer can add multiple user accounts. The User entity contains attributes such as username (primary key), email, and password, representing individuals who interact with the system.

Users are linked to the AddAgroProducts entity through the "can insert" relationship, which is many-to-many (M: N). This indicates that multiple users can add various agricultural products to the system. The AddAgroProducts entity has attributes such as pid (primary key), productname, productdesc, and price, representing details about the products.

The Farming entity stores information about farming types, with fid as the primary key and farmingtype as an attribute. It connects to the Trig entity, which logs actions or events triggered in the system, such as product-related updates. The Trig entity includes id (primary key), fid (foreign key), action, and timestamp as attributes, linking it to both the farming type and the triggered actions.

This system is designed to facilitate efficient management of farmer data, user accounts, agricultural products, and farming-related events while maintaining structured relationships among entities.

4.3 SCHEMA DIAGRAM:

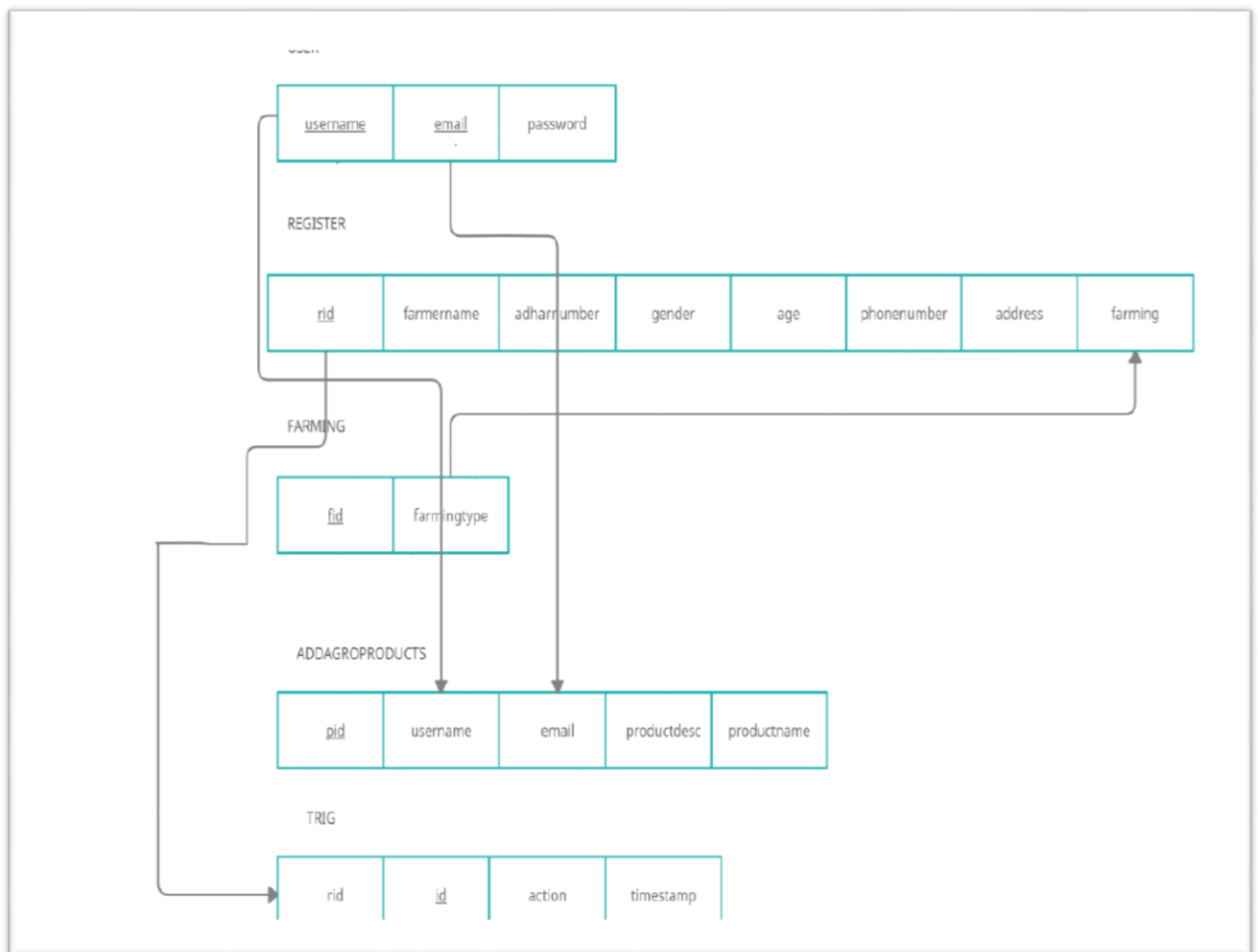


Fig 4.3: Schema Diagram

The provided schema diagram illustrates a relational database structure designed to manage data for users, farmers, farming activities, agricultural products, and triggered events. At its core, the USER table holds essential login credentials, including the username, email, and password, enabling a basic authentication system for users to register or log in. The REGISTER table expands on this by storing comprehensive details about farmers, such as their unique identifier (rid), name (farmername), Aadhar number (adharnumber), gender, age, phone number, address, and the type of farming they engage in, which links to the FARMING table. This integration allows for an organized repository of farmers' personal and professional details.

The FARMING table captures the various farming activities undertaken by farmers. It includes a unique farming identifier (fid) and a description of the farming type, such as crop cultivation or livestock farming, enabling the system to categorize farming practices. Additionally, the ADDAGROPRODUCTS table manages records of agricultural products added by users, with details like the product ID (pid), the user's username and email, as well as the product name (productname) and description (productdesc). This table facilitates tracking of agro-products and links them directly to the respective users who listed them.

Lastly, the TRIG table serves to monitor system actions and events. It includes fields for the unique identifier (id), action description, timestamp, and a link to the rid from the REGISTER table. This table ensures a robust logging mechanism to track and manage specific actions or events related to farmers. The relationships across these tables are clearly defined: the USER table connects to other tables via username and email, while the REGISTER table links to the FARMING and TRIG tables using the farmer ID (rid). Similarly, the ADDAGROPRODUCTS table relates to the USER table via username and email. Overall, this schema is thoughtfully designed for a farming or agriculture-based management system, ensuring structured and interconnected data storage for managing users, farmers, farming practices, products, and system events.

4.4 IMPLEMENTATION:

An "implementation" of Python should be taken to mean a program or environment which provides support for the execution of programs written in the Python language, as represented by the CPython reference implementation. There have been and are several distinct software packages providing of what we all recognize as Python, although some of those are more like distributions or variants of some existing implementation than a completely new implementation of the language.

❖ Back End (MySQL)

✓ Database:

A Database Management System (DBMS) is computer software designed for the purpose of managing databases, a large set of structured data, and run operations on the data requested by numerous users. Typical examples of DBMSs include Oracle, DB2, Microsoft Access, Microsoft SQL Server, Firebird, PostgreSQL, MySQL, SQLite, FileMaker and Sybase Adaptive Server Enterprise. DBMSs are typically used by Database administrators in the creation of Database systems. Typical examples of DBMS use include accounting, human resources and customer support systems. Originally found only in large companies with the computer hardware needed to support large data sets, DBMSs have more recently emerged as a fairly standard part of any company back office. DBMS is a complex set of software programs that controls the organization, storage, management, and retrieval of data in a database. A DBMS includes:

- A modelling language to define the schema of each database hosted in the DBMS, according to the DBMS data model.
- The dominant model in use today is the ad hoc one embedded in SQL, despite the objections of purists who believe this model is a corruption of the relational model, since it violates several of its fundamental principles for the sake of practicality and performance. Many DBMSs also support the Open Database Connectivity API that supports a standard way for programmers to access the DBMS.

Data structures (fields, records, files and objects) optimized to deal with very large amounts of data stored on a permanent data storage device (which implies relatively slow access compared to volatile main memory). A database query language and report

writer to allow users to interactively interrogate the database, analyse its data and update it according to the users' privileges on data.

- Data security prevents unauthorized users from viewing or updating the database. Using passwords, users are allowed access to the entire database or subsets of it called sub schemas. For example, an employee database can contain all the data about an individual employee, but one group of users may be authorized to view only payroll data, while others are allowed access to only work history and student data.
- If the DBMS provides a way to interactively enter and update the database, as well as interrogate it, this capability allows for managing personal databases. However, it may not leave an audit trail of actions or provide the kinds of controls necessary in a multi-user organization. These controls are only available when a set of application programs are customized for each data entry and updating function.
- ✓ A transaction mechanism, that ideally would guarantee the ACID properties, in order to ensure data integrity, despite concurrent user accesses (concurrency control), and faults (fault tolerance).
 - It also maintains the integrity of the data in the database.
 - The DBMS can maintain the integrity of the database by not allowing more than one user to update the same record at the same time. The DBMS can help prevent duplicate records via unique index constraints; for example, no two customers with the same customer numbers (key fields) can be entered into the database. See ACID properties for more information (Redundancy avoidance).

When a DBMS is used, information systems can be changed much more easily as the organization's information requirements change. to the Organizations may use one kind of DBMS for daily transaction processing and then move the detail onto another computer that uses

another DBMS better suited for random inquiries and analysis. Overall systems design decisions are performed by data administrators and systems analysts. Detailed database design is performed by database administrators.

✓ **SQL:**

Structured Query Language (SQL) is the language used to manipulate relational databases. SQL is tied very closely with the relational model.

- In the relational model, data is stored in structures called relations or tables.

SQL statements are issued for the purpose of:

- Data definition: Defining tables and structures in the database (DDL used to create, alter and drop schema objects such as tables and indexes).

CHAPTER 5

RESULT ANALYSIS AND SNAPSHOTS

In this chapter, Result analysis of the project results to a user-friendly Interface which is well aesthetic to avoid user's frustration in creating a resume. The Results consists of User Interface along with sign in, login, registration page, product details, triggers records, add farming, farmer details with the edit delete add option, adding agro products, along with a saved copy in database , and can even share through social platforms (email) .

5.1 USER INTERFACE SNAPSHOT:

1. Fig 5.1 Shows the snapshot of the main page:

The main page consists of Home, Farmer register, Add farming, farmer details, Agro products, Records and Sign in pages modules.

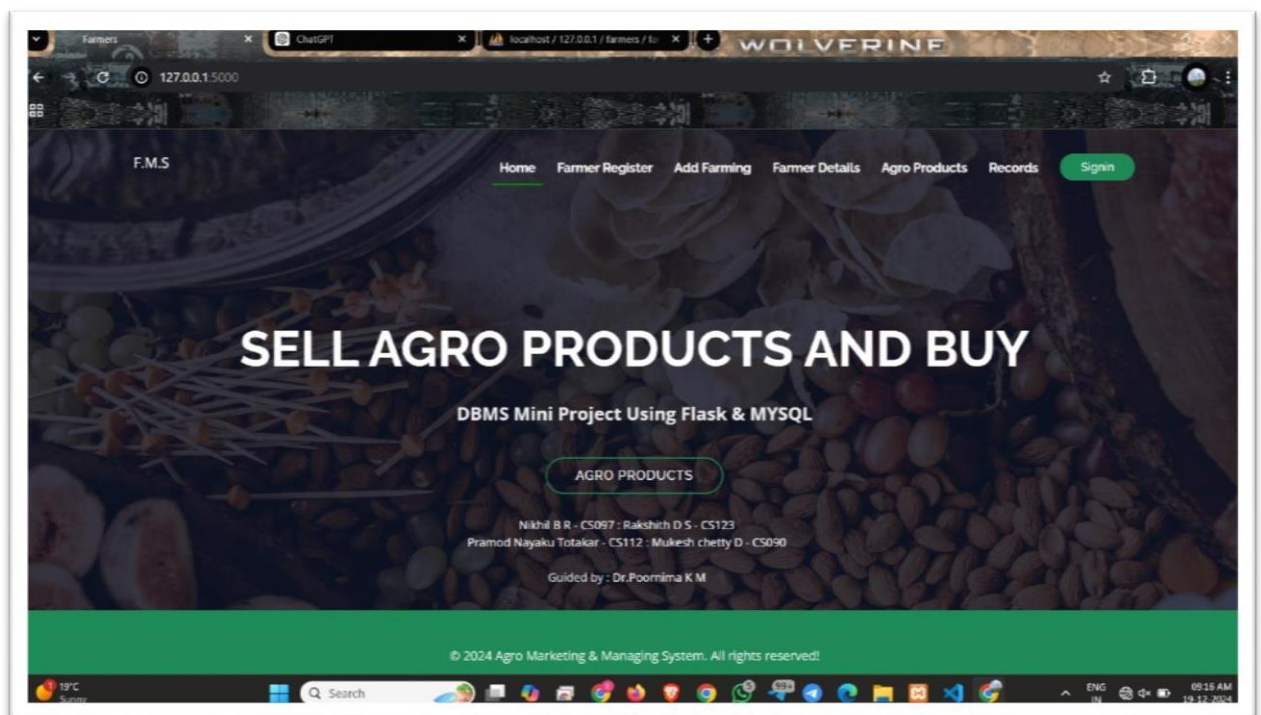


Fig 5.1: - Main Page

2. Fig 5.2 Shows the snapshot of the sign in page:

The sign in page consists of Username, Email address, Password, and a login page as a module. Sign up page leads to your username page.

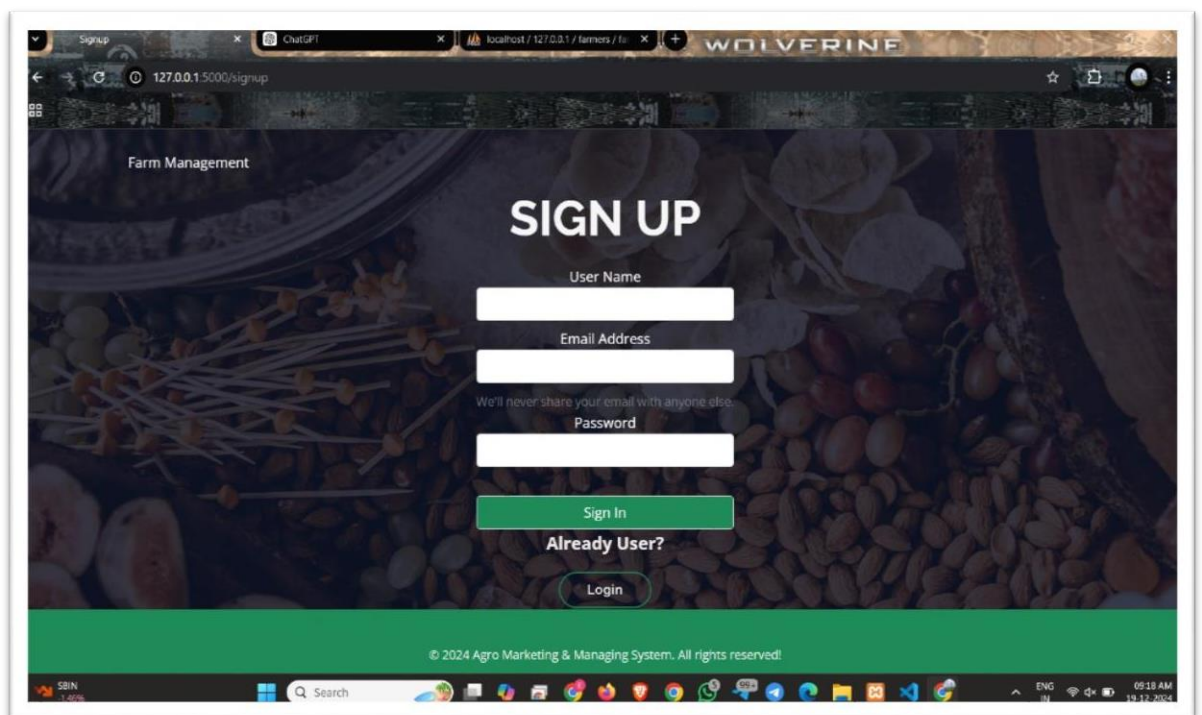


Fig 5.2: - Sign in Page

3. Fig 5.3 Shows the snapshot of the Products details page:

The Product details page consists of Agro products page as a module. Agro products page leads customers or a farmer to the purchase and seal the product.

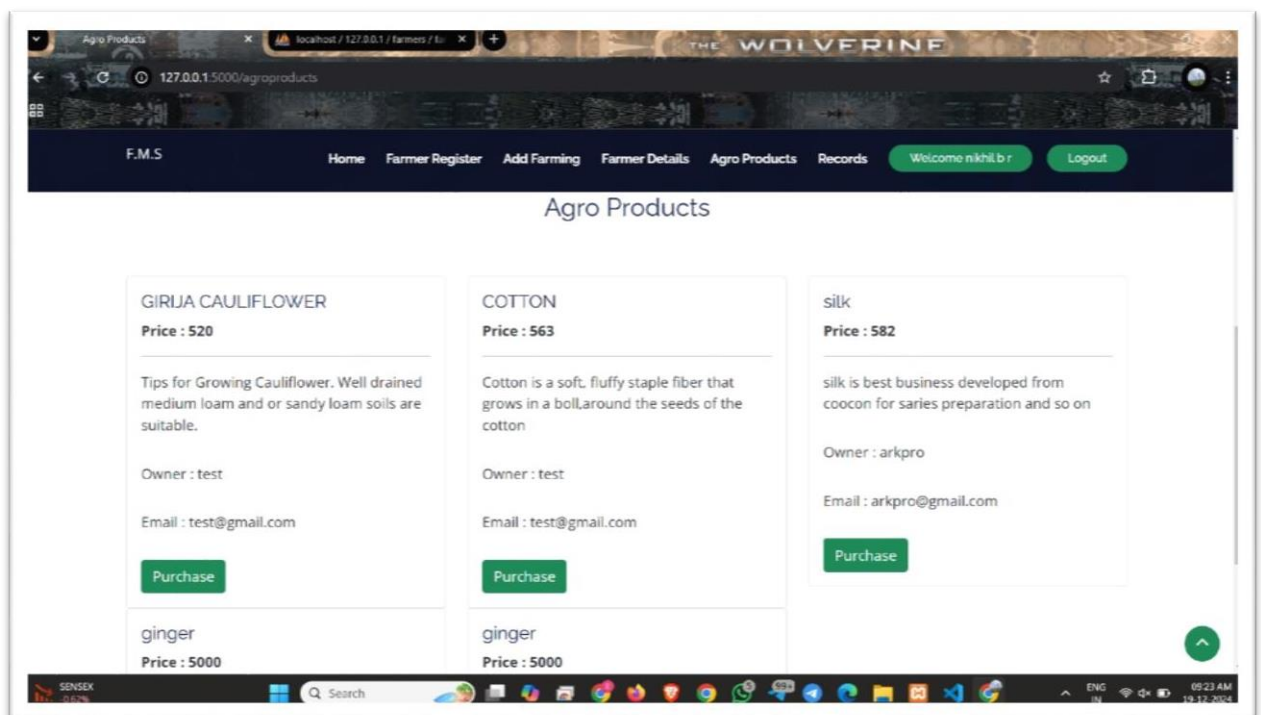


Fig 5.3: - Products details Page

4. Fig 5.4 Shows the snapshot of the E-mail page:

The E-mail page consists of an auto feed to email page as a module. Agro products page leads customers or a farmer to purchase and sale the product through the email, if a purchaser click on the purchase button it directly leads to email ,there we can type the required message to the farmer to communicate.

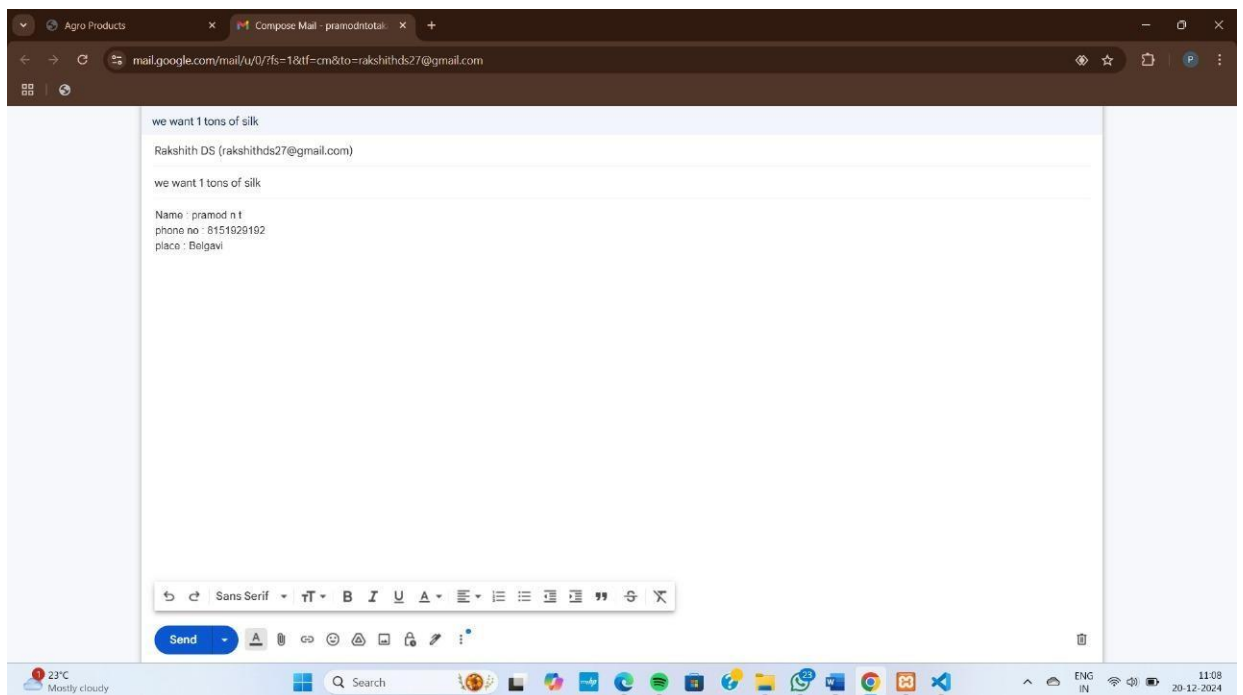


Fig 5.4: - E-mail Page

5. Fig 5.5 Shows the snapshot of the E-mail page in Mobile:

The email page in mobile displays the email sent by the purchaser to the farmer mobile to purchase the required product.

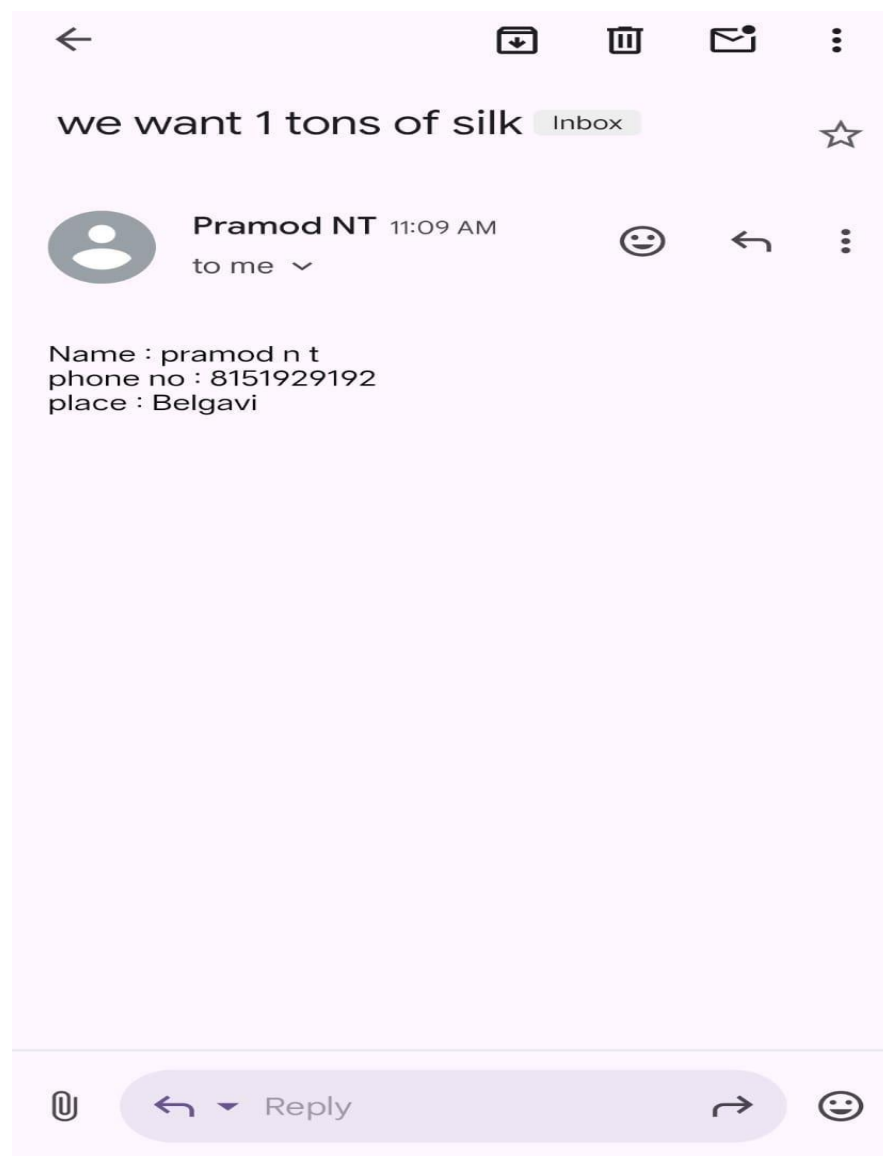
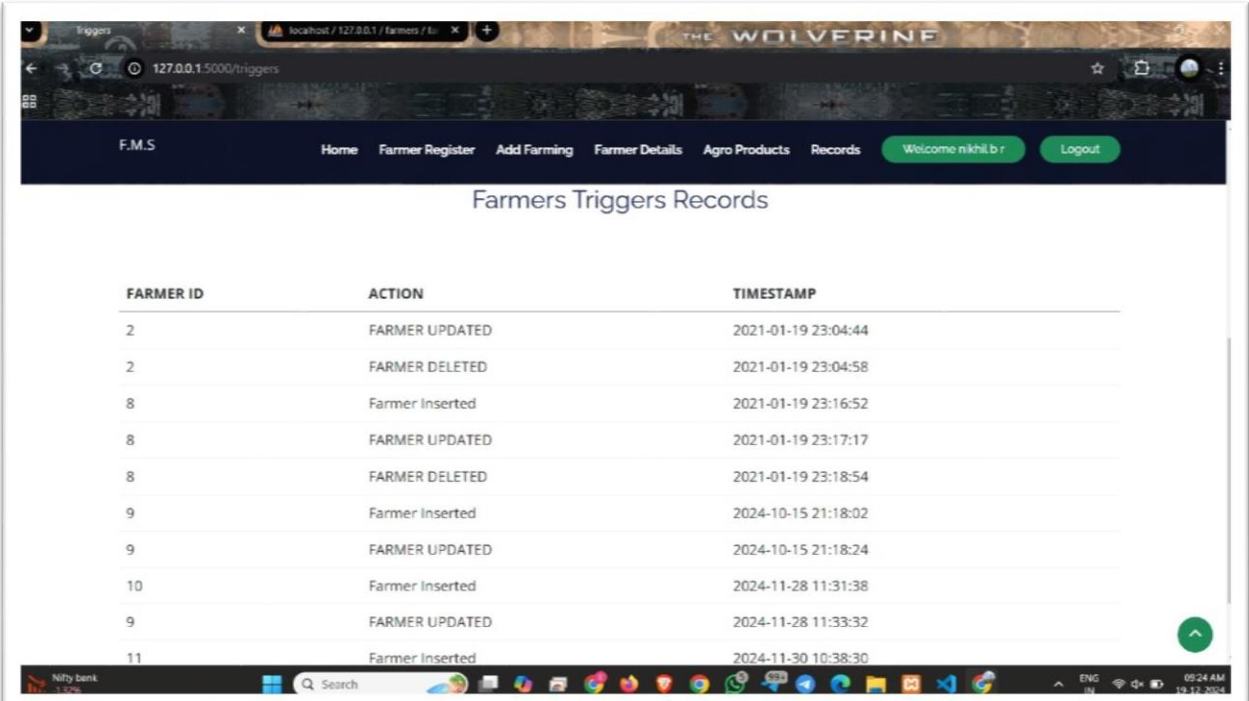


Fig 5.5: - E-mail Page in mobile

6. Fig 5.6 Shows the snapshot of the Triggers records page:

The Triggers Records page consists of Farmers Triggers Records page as a module, which leads customer or a farmer to the track their activity.



The screenshot shows a web application interface for 'Farmers Triggers Records'. The page has a dark blue header with navigation links: Home, Farmer Register, Add Farming, Farmer Details, Agro Products, and Records. A user is logged in as 'Welcome nikhil b r' with a 'Logout' button. The main content area displays a table with three columns: FARMER ID, ACTION, and TIMESTAMP. The table lists several records of farmer activities, including updates, deletions, and insertions, with timestamps ranging from 2021-01-19 to 2024-11-30. A green circular button with an upward arrow is visible in the bottom right corner of the table area.

FARMER ID	ACTION	TIMESTAMP
2	FARMER UPDATED	2021-01-19 23:04:44
2	FARMER DELETED	2021-01-19 23:04:58
8	Farmer Inserted	2021-01-19 23:16:52
8	FARMER UPDATED	2021-01-19 23:17:17
8	FARMER DELETED	2021-01-19 23:18:54
9	Farmer Inserted	2024-10-15 21:18:02
9	FARMER UPDATED	2024-10-15 21:18:24
10	Farmer Inserted	2024-11-28 11:31:38
9	FARMER UPDATED	2024-11-28 11:33:32
11	Farmer Inserted	2024-11-30 10:38:30

Fig 5.6: - Triggers records Page

7. Fig 5.7 Shows the snapshot of the Add farming page:

The Add farming page consists of adding products to purchase as a module, which leads customer or a farmer to add the product which they have grown.

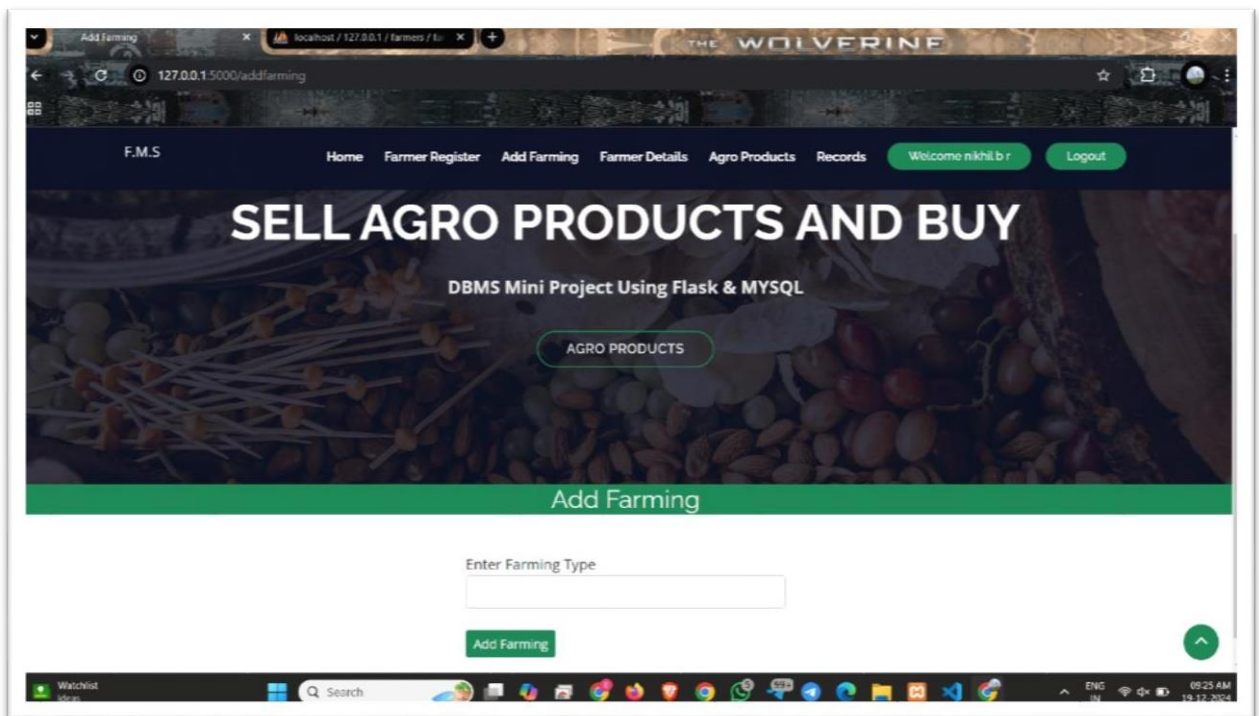


Fig 5.7: - Add farming Page

8. Fig 5.8 Shows the snapshot of the Farmer Registration page:

The Farmer Registration page consists of Farmer name, Adhar number, Age, Gender, Phone number, Address and the product name as a module, which leads customer or farmer to feed farmers details.

Register Farmers Details

Farmer Name

Adhar Number

Age

Gender

Phone Number

Address

Product Name

Fig 5.8: - Registration Page

9. Fig 5.9 Shows the snapshot of the Farmer details page:

The Farming details consists of Registered farmer details as a module, which leads customer or a farmer to add, delete, and edit the farmers' details.

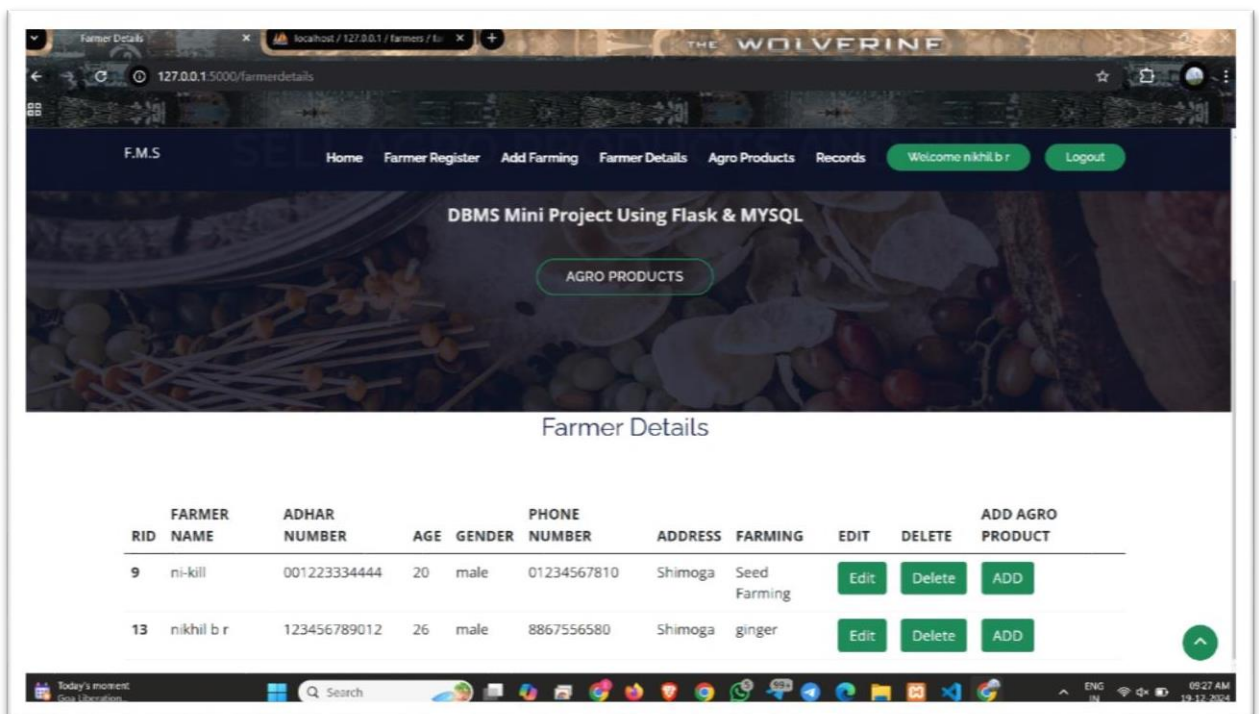


Fig 5.9: - Farmer details Page

10. Fig 5.10 Shows the snapshot of the Adding Agro products page:

The Adding Agro products details consists of Farmers name, Farmer email, Product name, Product description, Price as a module, which leads to add the product details to purchase.

The screenshot displays a web application interface for adding agro products. The header includes a navigation menu with options like 'Home', 'Farmer Register', 'Add Farming', 'Farmer Details', 'Agro Products', and 'Records'. A user is logged in as 'nikhil b r' with a 'Logout' button. The main form contains the following fields:

- Farmer Name: nikhil b r
- Farmer Email: nikkibalappar@gmail.com
- Product Name: ginger
- Product Description: we have best quality in the market, In tons.
- Price: 3500

An 'Add Product' button is located at the bottom of the form. The browser address bar shows '127.0.0.1:5000/addagroproduct'.

Fig 5.10: - Adding Agro products Page

5.2 DATE BASE SNAPSHOT:

In this date base snapshot, which shows the Register data base, Trigger data base, Add products data base and user data base. The respective information is feeded to the respective data base column.

11. Fig 5.11 Shows the snapshot of the Register Database page:

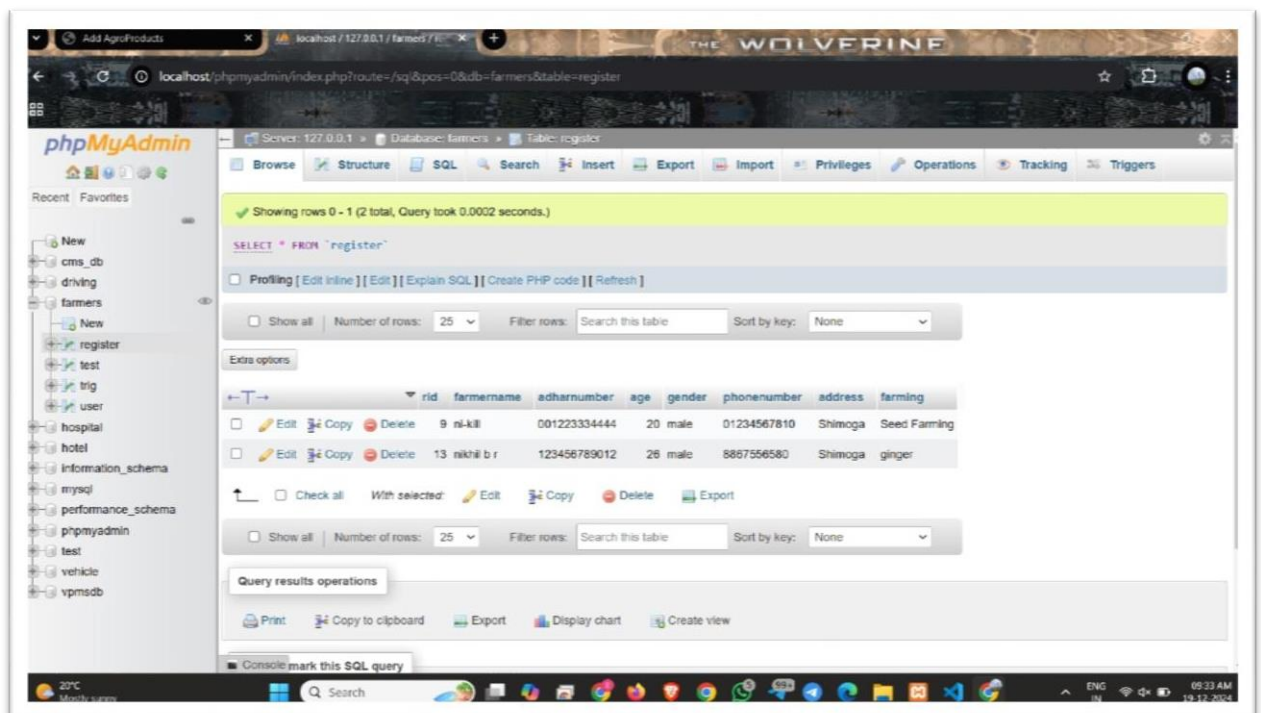


Fig 5.11: - Register Database Page

12. Fig 5.12 Shows the snapshot of the Trigger Database page:

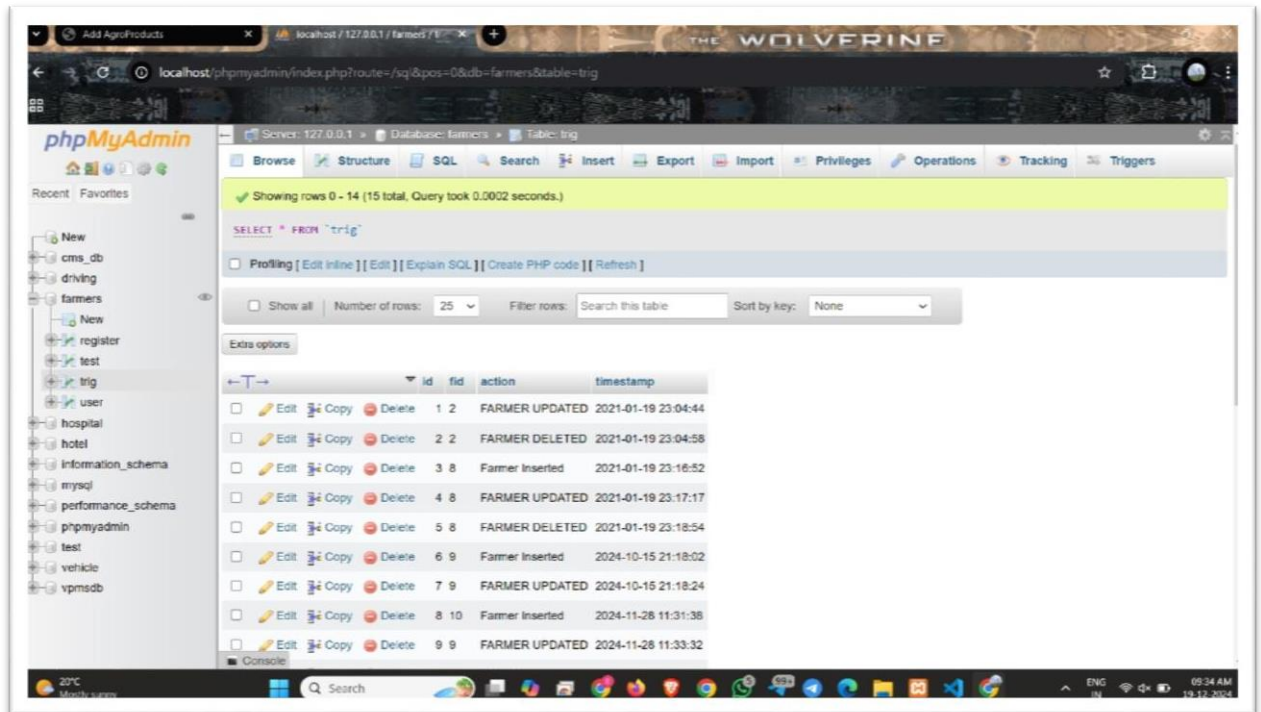


Fig 5.12: - Trigger Database Page

13. Fig 5.13 Shows the snapshot of the Add product Database page:

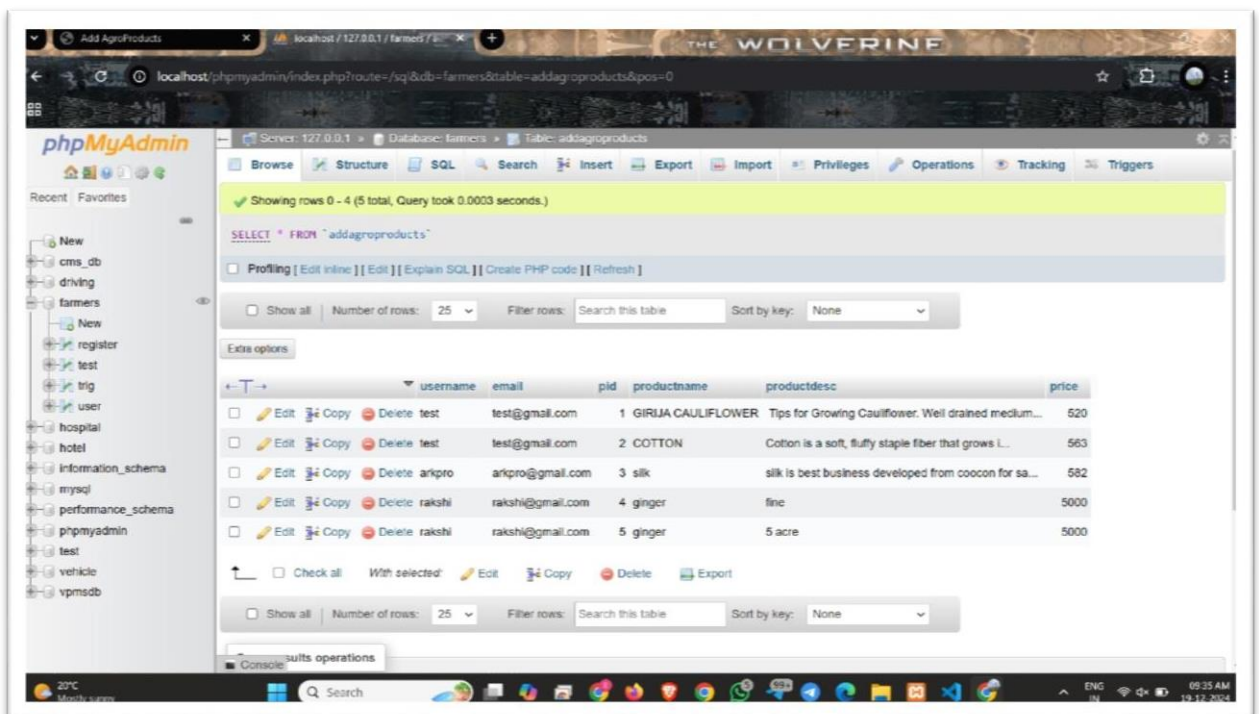


Fig 5.13: - Add product Database Page

14. Fig 5.14 Shows the snapshot of the User Database page:

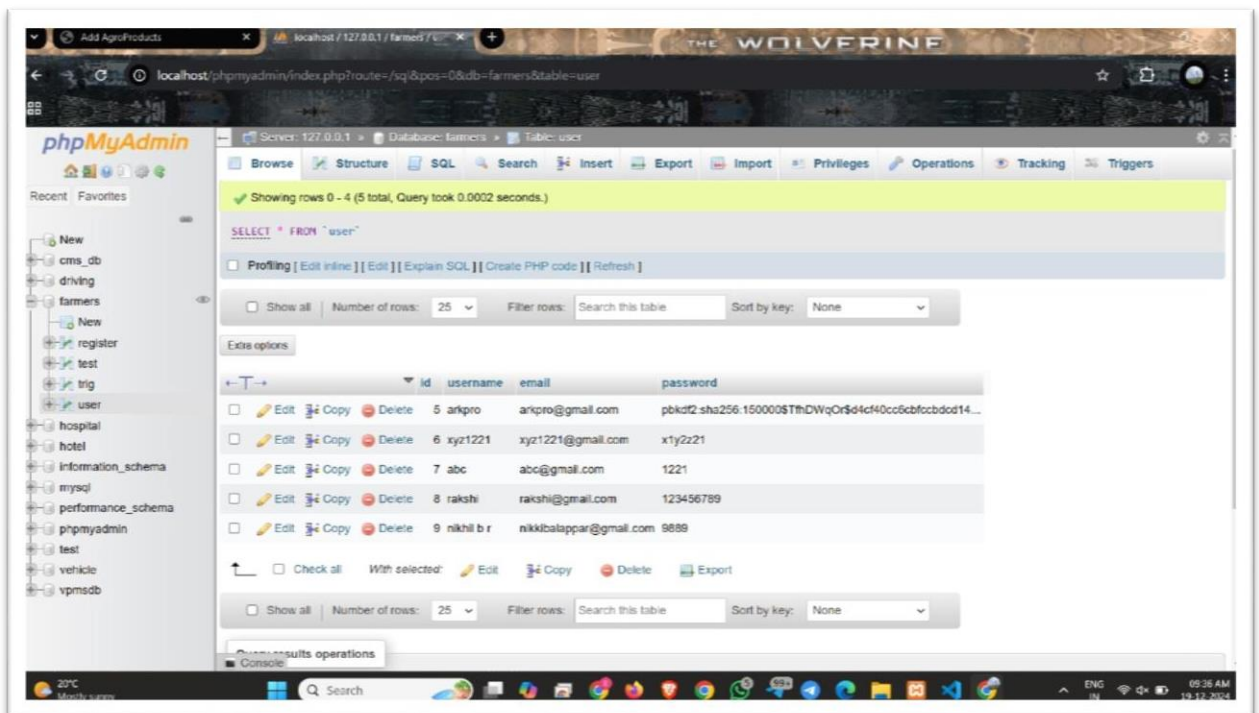


Fig 5.14: - User Database Page

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The **Agro Marketing and Management System (AMMS)** represents a significant advancement in the agricultural sector, offering a comprehensive solution to address several challenges faced by farmers, consumers, and other stakeholders in the supply chain. By digitizing the entire process from production to sales, AMMS empowers farmers by providing them with a direct and transparent connection to consumers, reducing the dependency on intermediaries. This direct market access allows farmers to achieve better pricing for their products, increasing their profitability and financial sustainability. This system not only empowers farmers by providing them with essential tools to efficiently market their produce but also enhances transparency and trust among stakeholders. With its potential to reduce wastage, optimize supply chains, and ensure fair pricing, the AMMS contributes to sustainable agricultural practices and supports economic growth in rural communities.

6.2 FUTURE SCOPE FOR THE “AAMS”

The **Agro Marketing and Management System (AAMS)** has a promising future, with significant potential to revolutionize agriculture through technological advancements and expanded services. The integration of technologies like **Artificial Intelligence (AI)** can provide predictive analytics for crop demand, price trends, and market behaviors, enabling farmers to make data-driven decisions. Similarly, **Internet of Things (IoT)** devices can enable real-time monitoring of crop health, soil conditions, and weather, improving farm productivity. **Blockchain technology** can be incorporated to ensure secure, transparent transactions and product traceability, building trust between farmers and consumers. In the long term, AAMS could facilitate global market integration, helping farmers expand their reach through export opportunities and certifications like fair trade. By addressing challenges such as digital literacy and access to technology, AAMS can evolve into a vital tool for creating a more sustainable, efficient, and equitable agricultural ecosystem.

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Reference link:

<https://youtu.be/eQkLD5b5m4I?si=pSQwU-MNlgWyKhmP>